

# NumPy

## Summations

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## Summations

What is the difference between summation and addition?

Addition is done between two arguments whereas summation happens over n elements.

### Example

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Add the values in arr1 to the values in arr2:

```
import numpy as np

arr1 = np.array([1, 2, 3])
arr2 = np.array([1, 2, 3])

newarr = np.add(arr1, arr2)

print(newarr)
```

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Sum the values in arr1 and the values in arr2:



## NumPy Editor

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**Abstract**

```
import numpy as np

arr1 = np.array([1, 2, 3])
arr2 = np.array([1, 2, 3])
```

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```
newarr = np.sum([arr1, arr2], axis=1)  
  
print(newarr)
```

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**Returns:** `[6 6]`

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## Cummulative Sum

Cummulative sum means partially adding the elements in array.

E.g. The partial sum of [1, 2, 3, 4] would be [1, 1+2, 1+2+3, 1+2+3+4] = [1, 3, 6, 10].

Perform partial sum with the `cumsum()` function.

## Example

Perform cummulative summation in the following array:

```
import numpy as np  
  
arr = np.array([1, 2, 3])  
  
newarr = np.cumsum(arr)  
  
print(newarr)
```

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**Returns:** `[1 3 6]`